The Fourier Qualia Space:

Interaction, Ambiguity, and Hierarchy in Music Harmony

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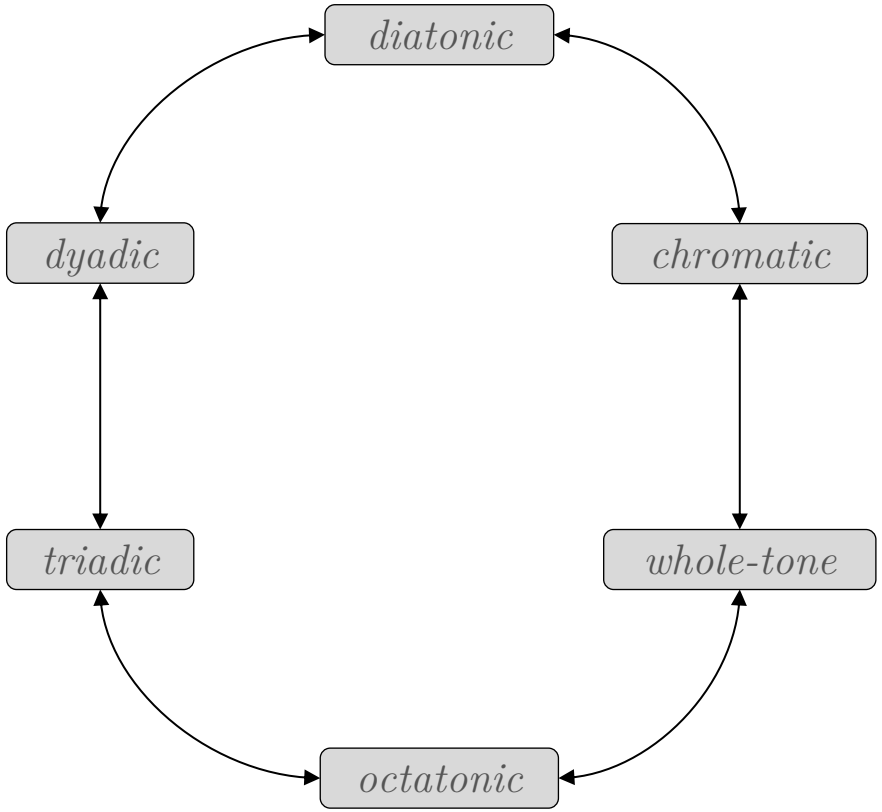




I. Prelude.

Preliminary concepts, context, and the catch.

1. What a harmonic quality is.



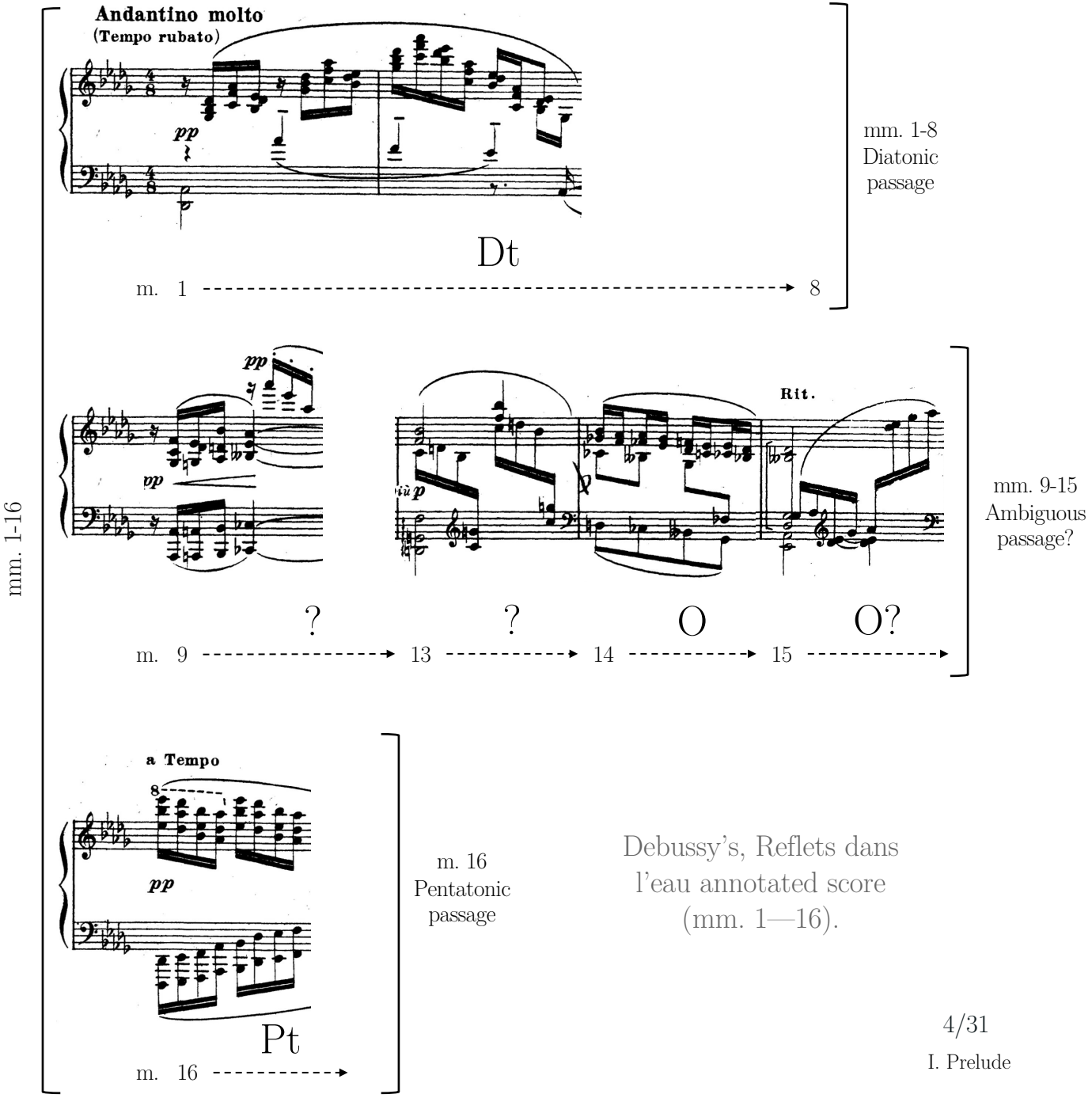
Harmonic qualities: heard directly, composed with deliberately.

A harmonic quality is the colour of the collection a passage is drawn from. It is heard without identifying a specific key, and unchanged by transposition or revoicing.

2. An example.

Listen.
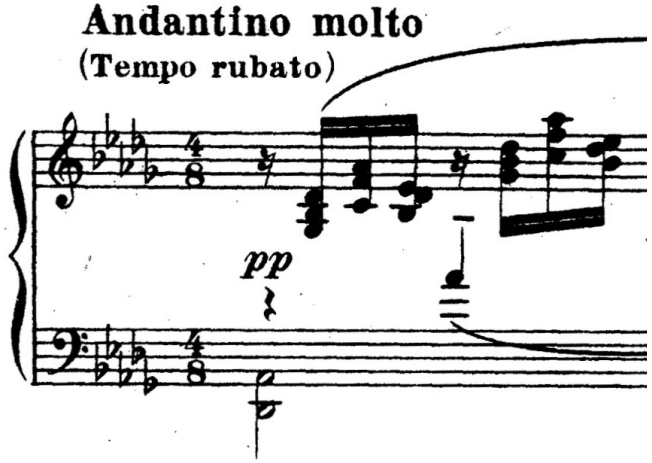
Debussy, Reflets dans l'eau
mm. 1—16. Listen 1/2.



Debussy's, Reflets dans l'eau annotated score (mm. 1—16).

3. A few questions.

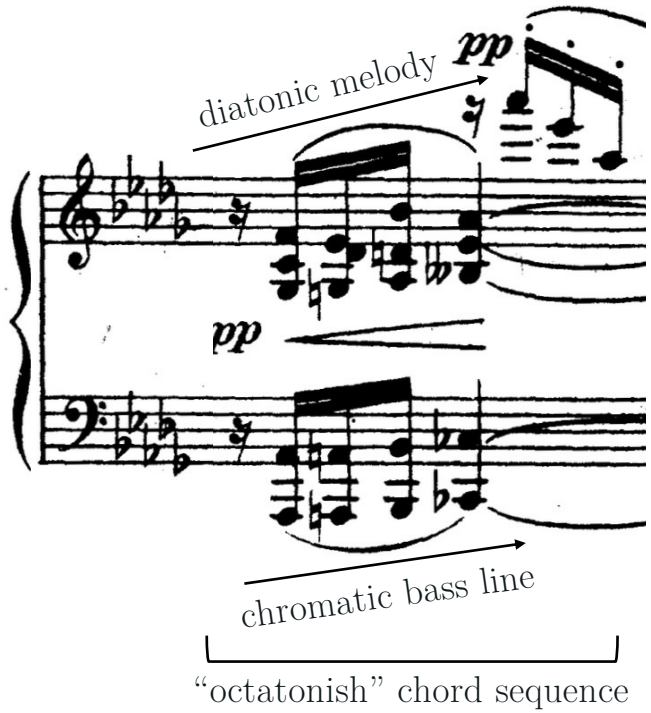
1. How diatonic was m. 1?



3. A few questions.

1. How diatonic was m. 1?

2. How quality-ambiguous did m. 9 become?

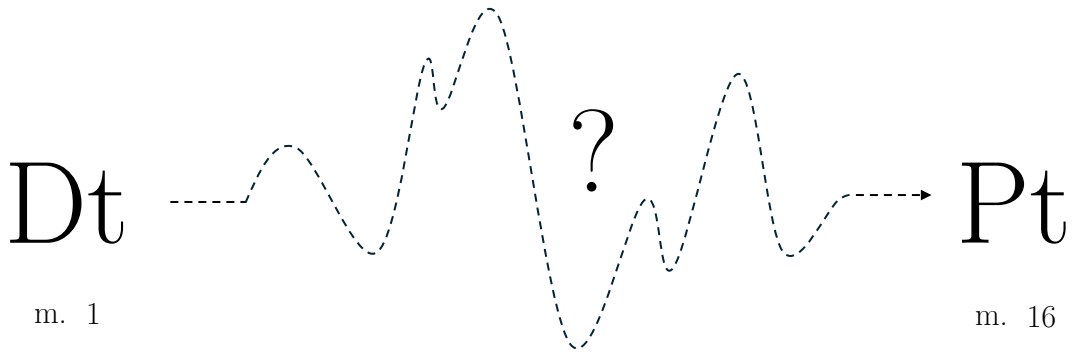


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1. How diatonic was m. 1?

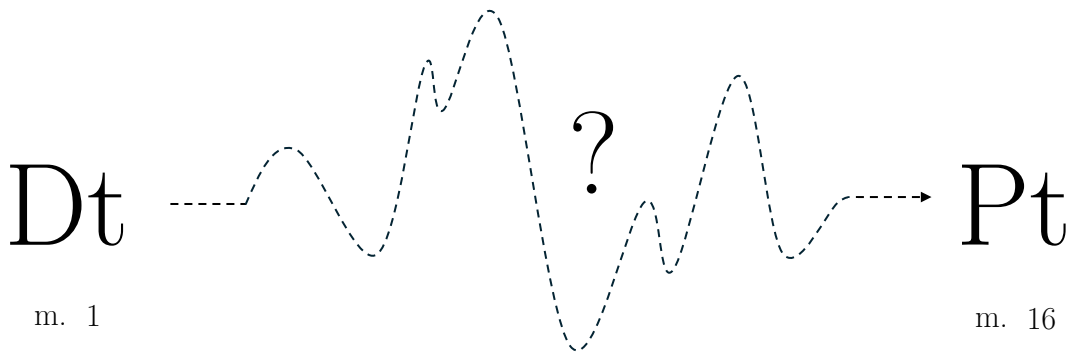
2. How quality-ambiguous did m. 9 become?

3. What path did the music take between them?



3. A few questions.

1. How diatonic was m. 1?
2. How quality-ambiguous did m. 9 become?
3. What path did the music take between them?



> **Central enquiry:** Can analysis of quality deployment expose systematic principles governing how qualities succeed one another in time?

4. Qualia: why the word.

> Harmonic-quality reconceived across four dimensions:
>
> 1. Spatial-geometric: quality as position in space (Quinn, 2006–7);
> 2. Temporal-accumulative: macroharmony (Tymoczko, 2011);
> 3. Hierarchical: quality across structural levels (Viaccoz et al., 2023);
> 4. Phenomenological: the measurements correspond to something we perceive.

Note: not a perceptual study. We argue that what the Fourier transform measures carries perceptual implications, but these are not tested experimentally here.

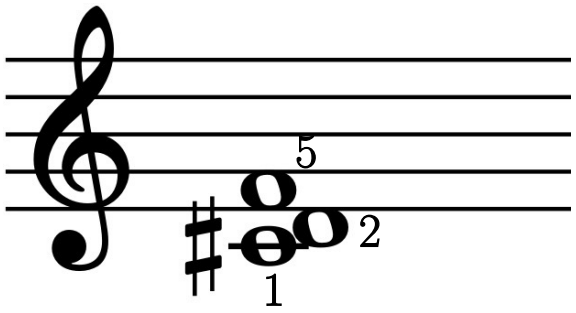
II. Proposition.

Foundations and the Fourier Qualia Space.

1. Pitch-class sets as binary vectors.

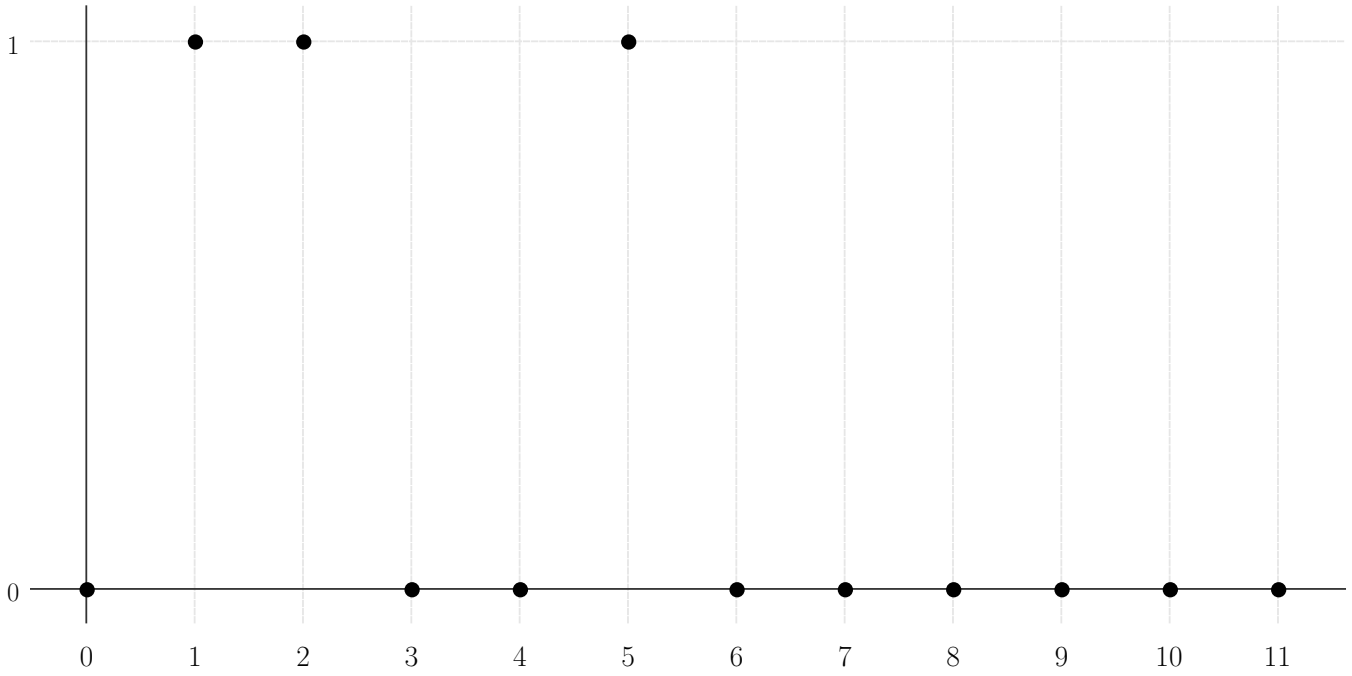
- Every pitch-class set becomes a 12-element binary vector over the pitch-classes 0–11.
- Example:

  {C♯, D, F} →
  {1, 2, 5} →
  (0,1,1,0,0,1,0,0,0,0,0,0).



Pitch-class set {1, 2, 5}.

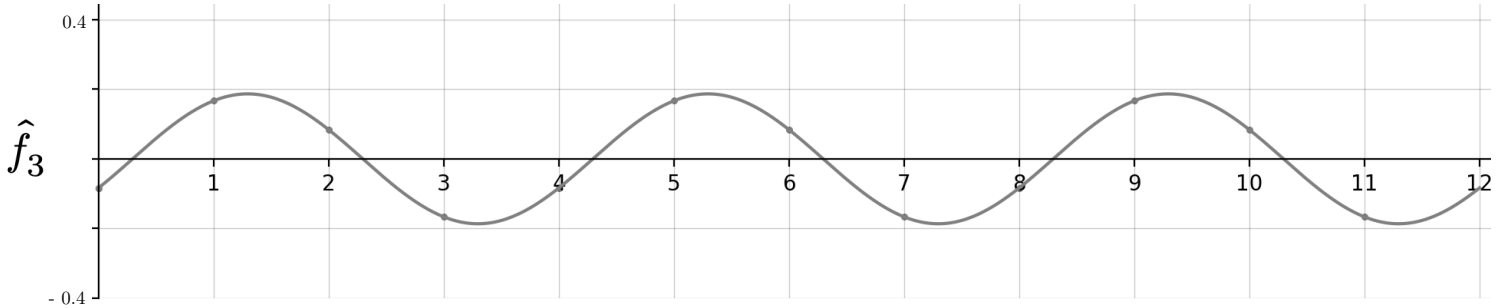
- Minimally abstract—a concrete object.
- This representation is the input for every step that follows.



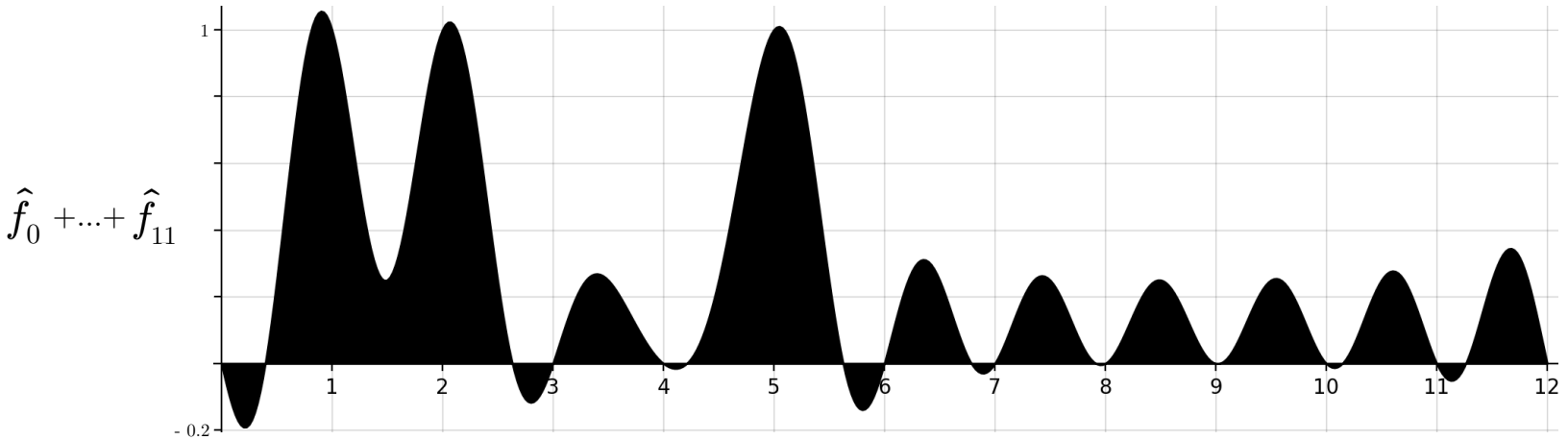
Binary encoding of pitch-class set {1, 2, 5} as a 12-element vector.

2. The Fourier transform of that vector.

The DFT decomposes any pitch-class vector into **twelve coefficients** $f_{0—11}$. Each coefficient **captures a specific periodicity** in the pitch-class distribution.



Sinusoid of the DFT coefficient 3 calculated for pitch-class set {1, 2, 5}.

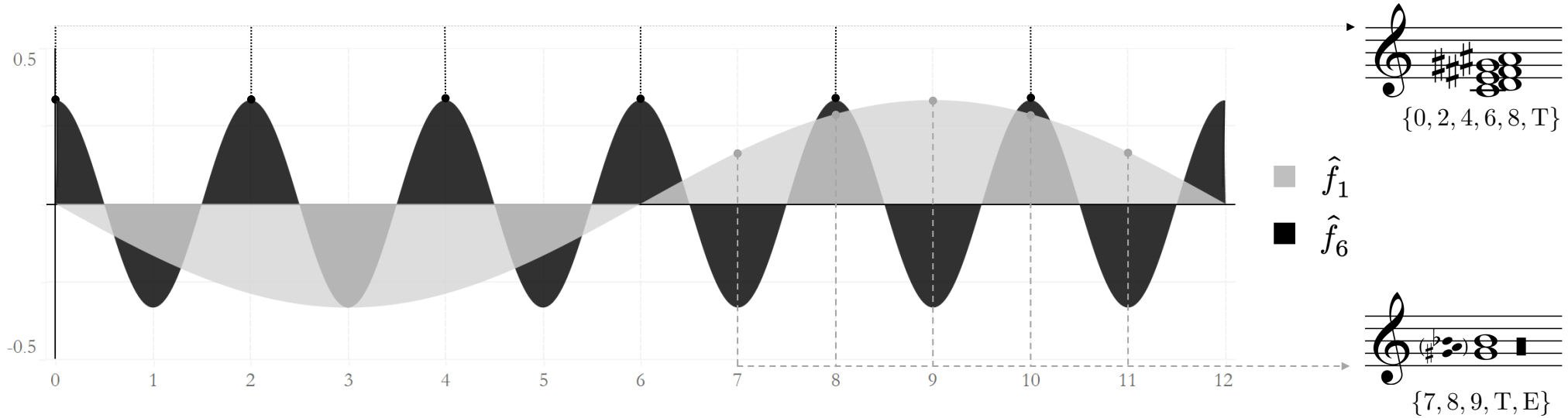


The sum of all twelve DFT coefficients reconstructs pitch-class set {1, 2, 5}.

3. Six non-trivial DFT coefficients, six harmonic qualities.

Six non-trivial coefficients map to six qualities:

- $\hat{f}_1 \rightarrow$ chromaticity (one period)
- $\hat{f}_6 \rightarrow$ whole-tone quality (six periods)

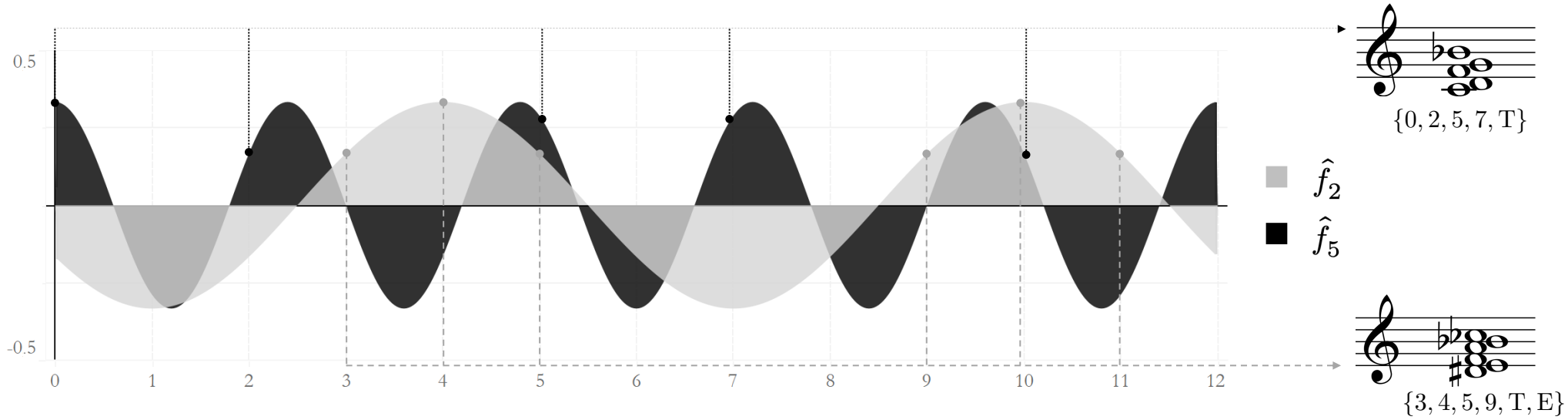


Two DFT coefficient sinusoids paired by period: chromaticity and whole-tone quality.

3. Six non-trivial DFT coefficients, six harmonic qualities.

Six non-trivial coefficients map to six qualities:

- $\hat{f}_1 \rightarrow$ chromaticity (one period)
- $\hat{f}_2 \rightarrow$ dyadicity (two periods)
- $\hat{f}_6 \rightarrow$ whole-tone quality (six periods)
- $\hat{f}_5 \rightarrow$ diatonicity (five periods)

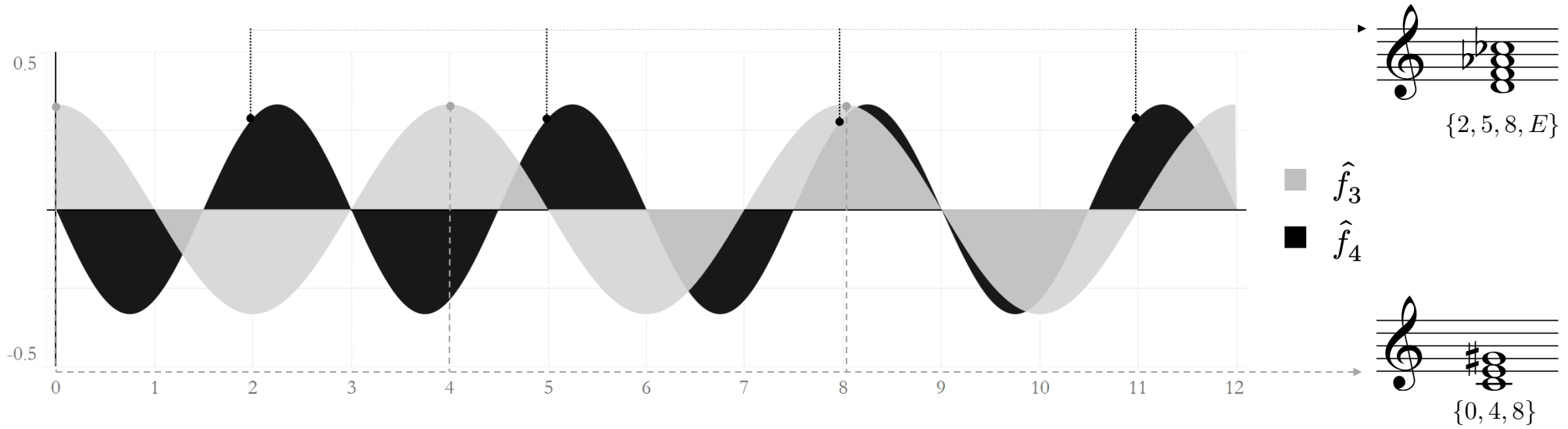


Two DFT coefficient sinusoids paired by period: dyadicity and diatonicity.

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Six non-trivial coefficients map to six qualities:

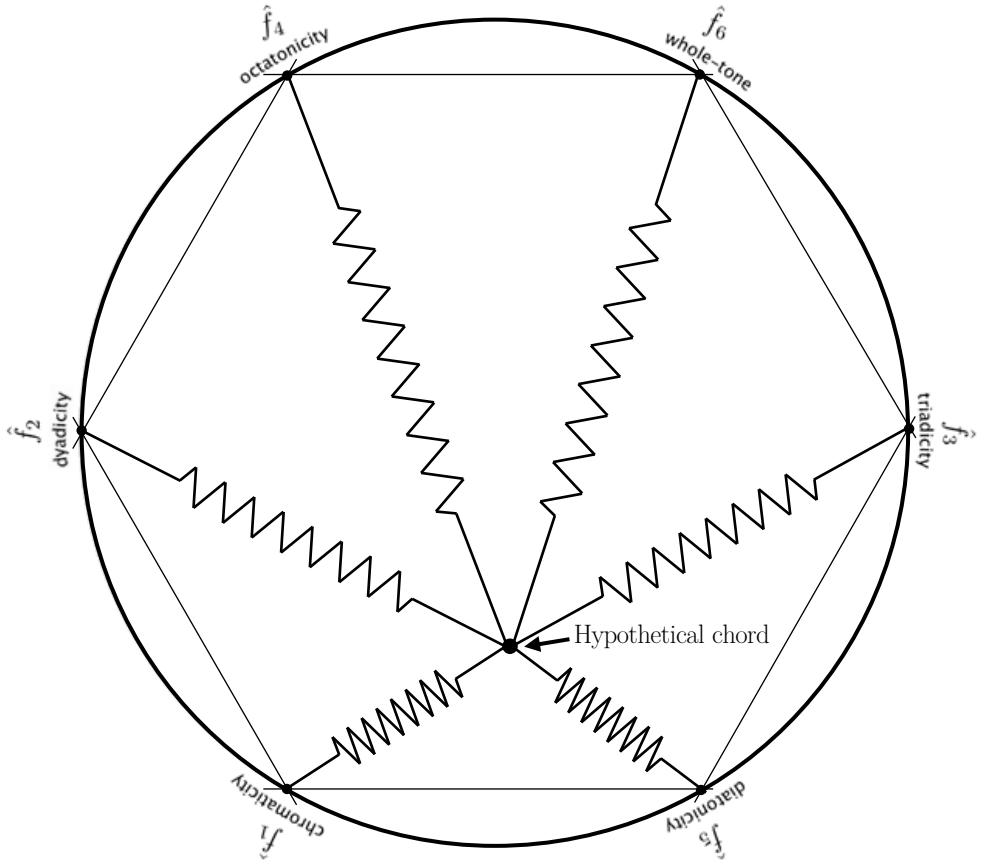
- $\hat{f}_1 \rightarrow$ chromaticity (one period)
- $\hat{f}_2 \rightarrow$ dyadicity (two periods)
- $\hat{f}_3 \rightarrow$ triadicity (three periods)
- $\hat{f}_6 \rightarrow$ whole-tone quality (six periods)
- $\hat{f}_5 \rightarrow$ diatonicity (five periods)
- $\hat{f}_4 \rightarrow$ octatonicity (four periods)



Two DFT coefficient sinusoids paired by period: triadicity and octatonicity.

4.1. Building the FQS: mechanical analogy.

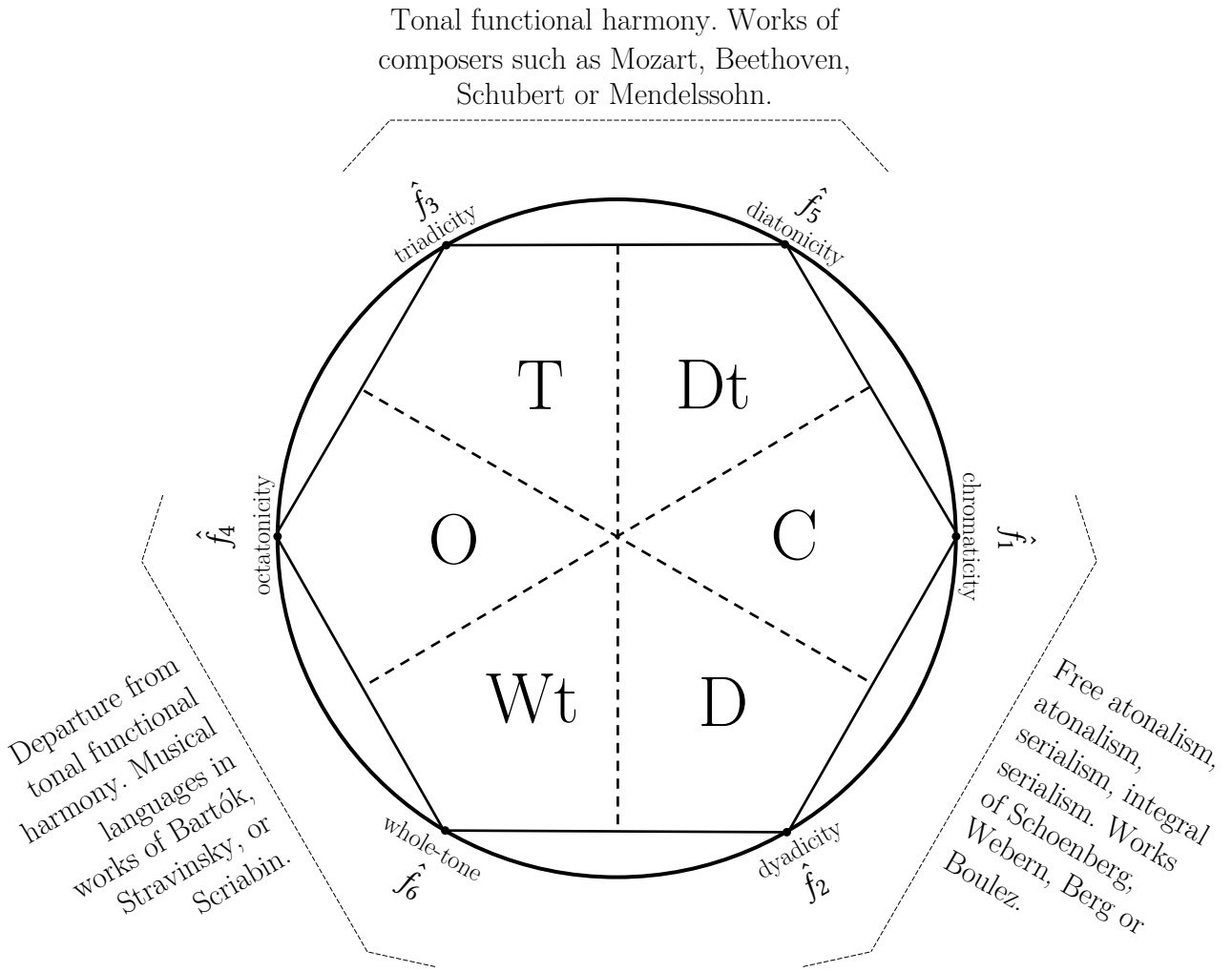
Picture a chord pulled by **six springs** toward the six hexagon anchors. Spring stiffness = DFT magnitude of the corresponding coefficient.



RadViz anchor positioning in the FQS and its spring model of forces.

4.2. Building the FQS: regions and meta-regions.

The FQS naturally partitions into **six primary regions**, one per quality, and spontaneously divides into three pairs of adjacent qualities.

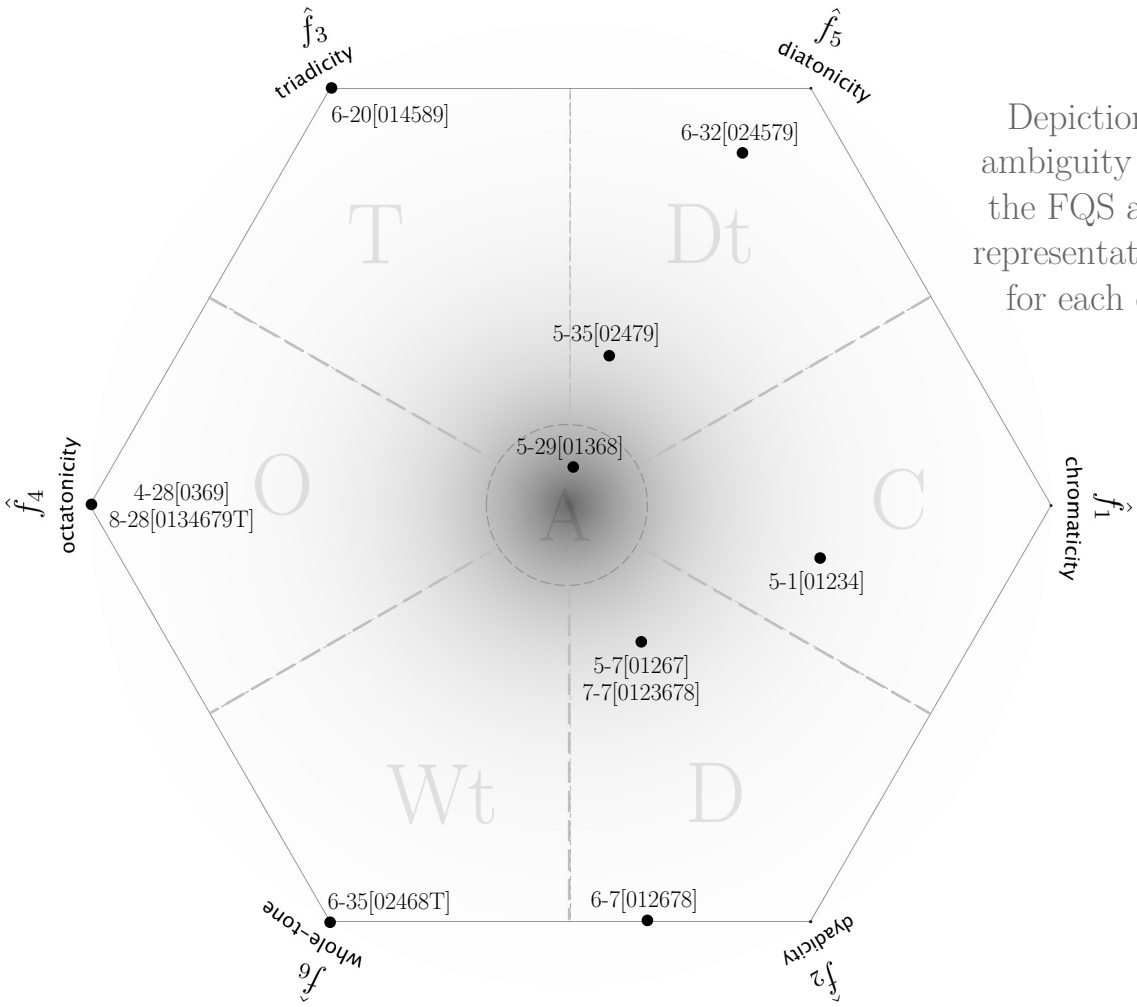


Primary regions and meta-regions over the FQS.

4.3. Building the FQS: qualia ambiguity (η) in the centre.

The FQS's centre stands as a singular point where no quality predominates:

- A seventh region thus denotes qualia ambiguity.
- η quantifies positional indefiniteness in the FQS.



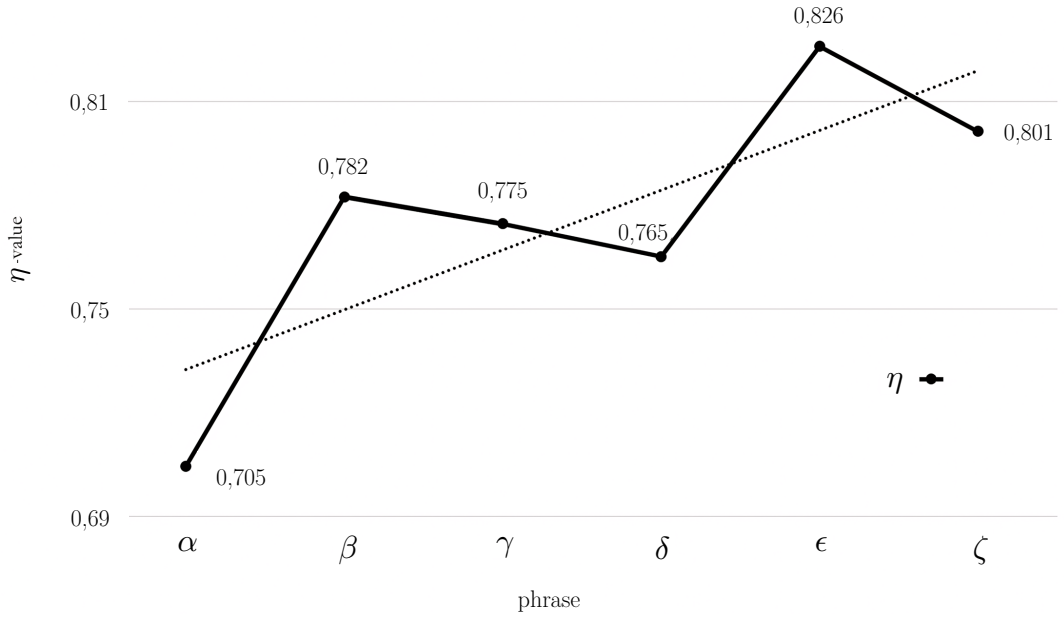
Depiction of the ambiguity region in the FQS and some representative pcsets for each quality.

III. What it has shown.

Contrapuntal ambiguity and hierarchical wavescapes.

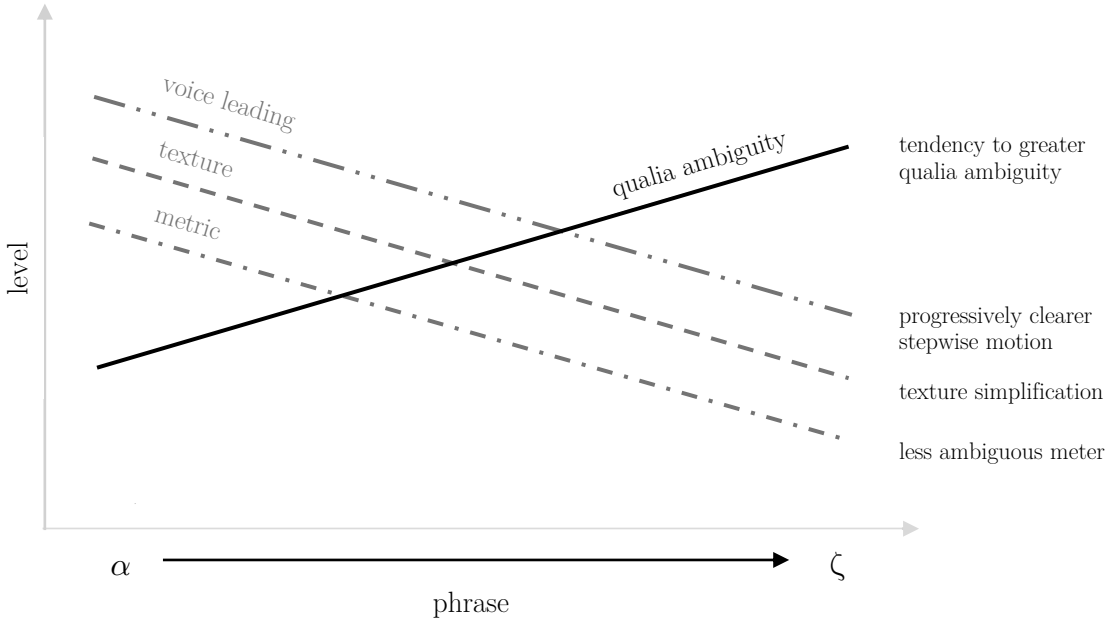
1. Schoenberg's Op. 19/1: contrapuntal linearities.

- Qualia ambiguity rises across the six phrases of Schoenberg's Op. 19/1.



Evolution of qualia ambiguity in Op. 19/1.

- It also behaves as an independent parameter, in counterpoint with others. *Music Theory Spectrum.*



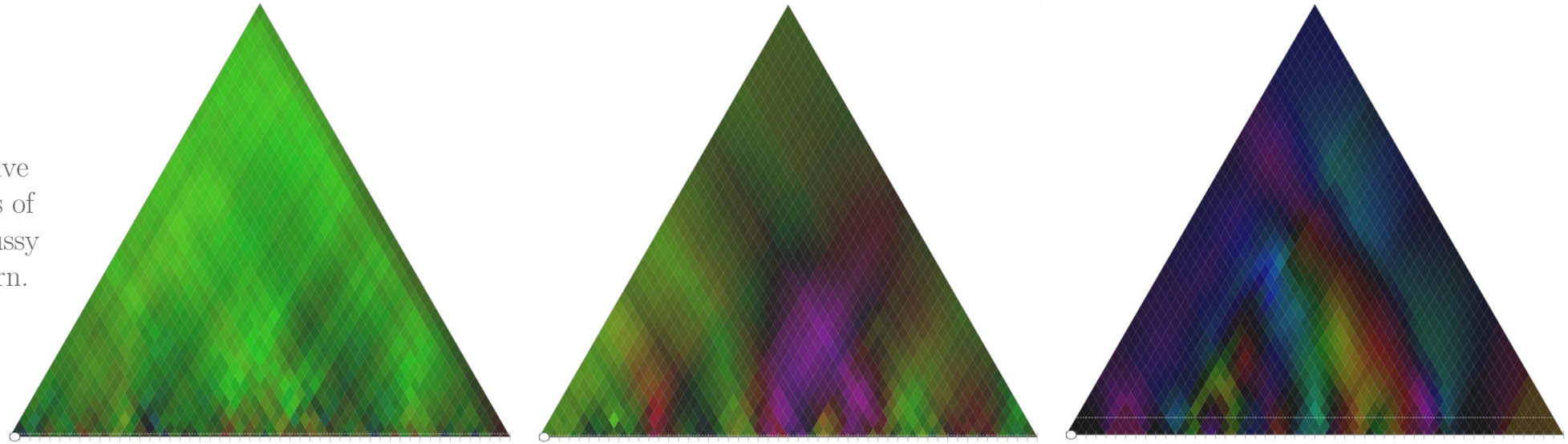
Nondirected and contrapuntal linearities in Op. 19/1.

2. Qualia wavescapes: three centuries, three strategies.

- Base = short segments, apex = the whole work; hue = harmonic quality, darkness = ambiguity.
- Bach green (diatonic) throughout; Debussy pendular (diatonic/whole-tone); Webern kaleidoscopic.
- *Mathematics and Computation in Music. midiVERTO app.*

Comparative wavescapes of Bach, Debussy and Webern.

- Increased qualia ambiguity?



- Bach's Three-Part Invention wavescape.
- Debussy's *Reflets dans l'eau* wavescape.
- Webern's Op. 7, No. 1 wavescape.

1723 — 1905 — 1910-1914

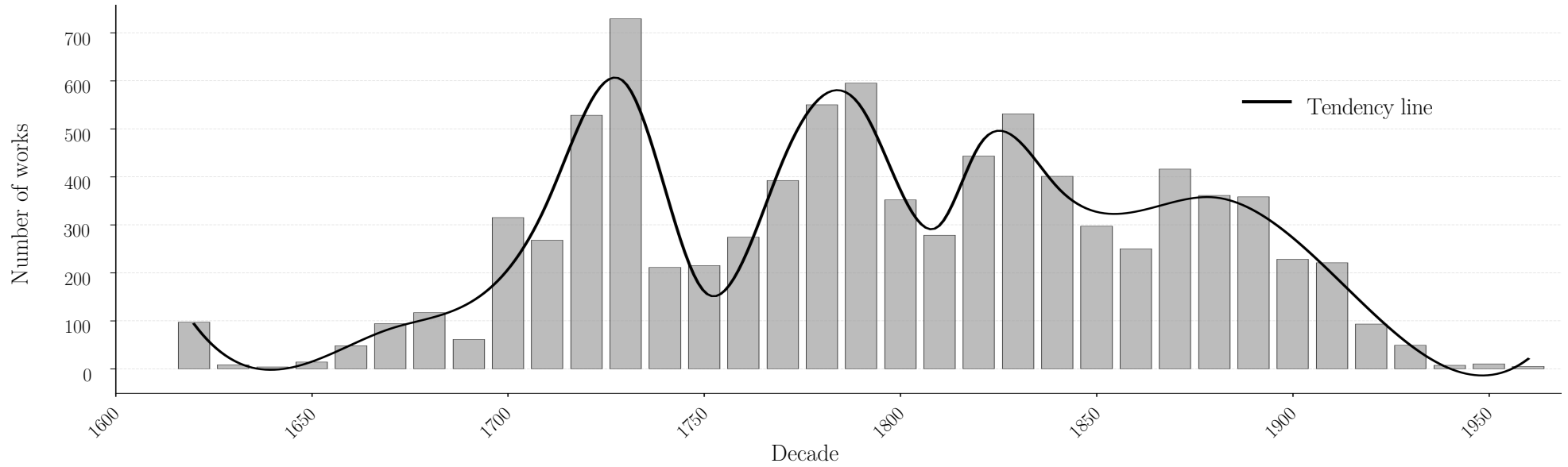
Timeline

IV. A statistical turn.

Diachronic analysis and Debussy's three role model.

1. The corpus.

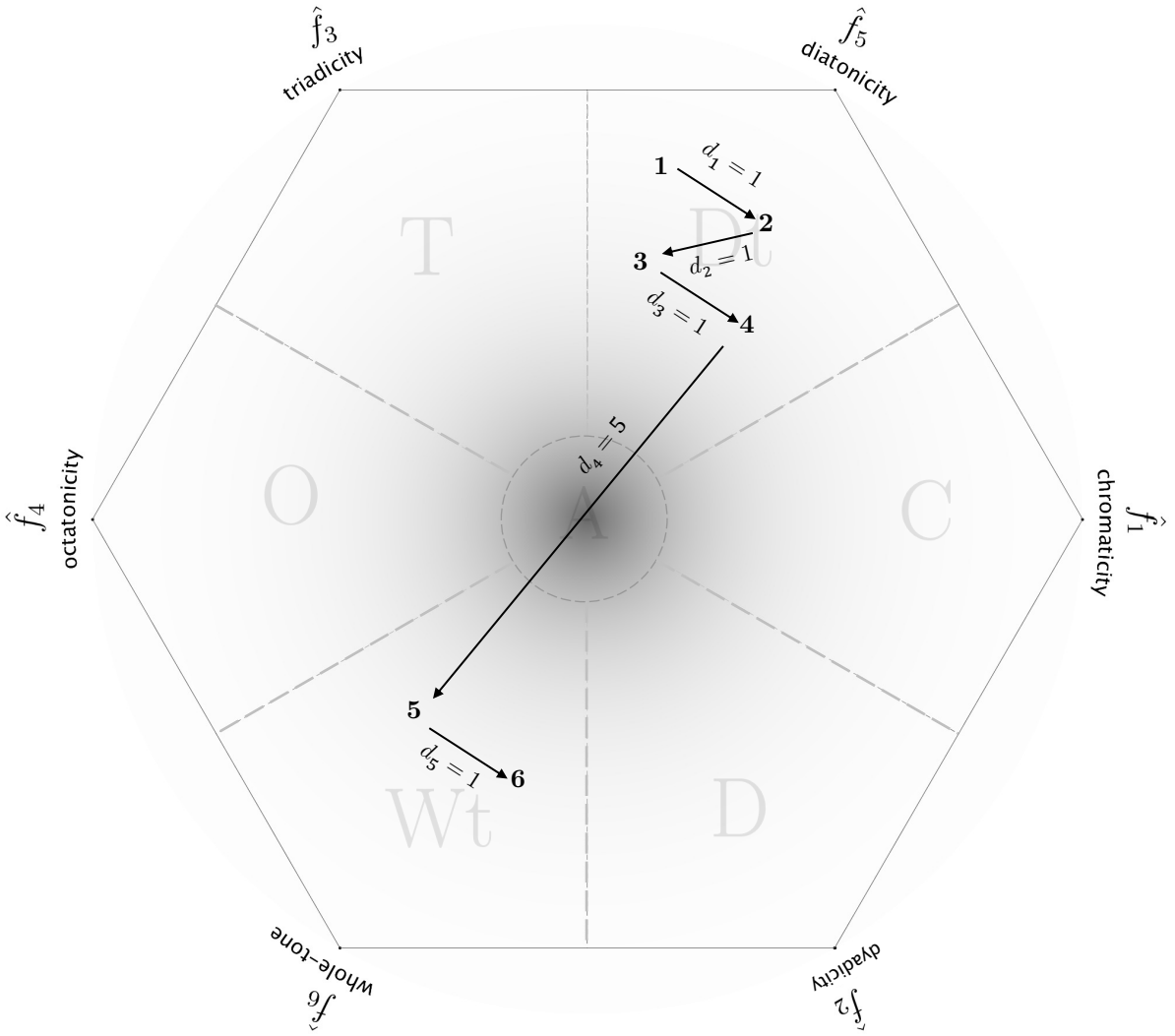
- Extended Yale–Classical Archives corpus.
- 8,822 works; 501 composers; 1548–1968.
- Augmented with 49 modernist works (Schoenberg 24, Webern 15, Messiaen 9, Berg 1) to address the post-1890 under-representation.



Distribution of works by decade in the extended corpus.

2.1. Adaptive segmentation: distance series for boundary detection.

- Every chord, for each piece in the corpus is plotted in the FQS.
- A distance-series is then calculated to find peaks and qualia change boundaries.
- d_4 is a peak in the distance series, indicating a change of quality between chords 4 and 5.



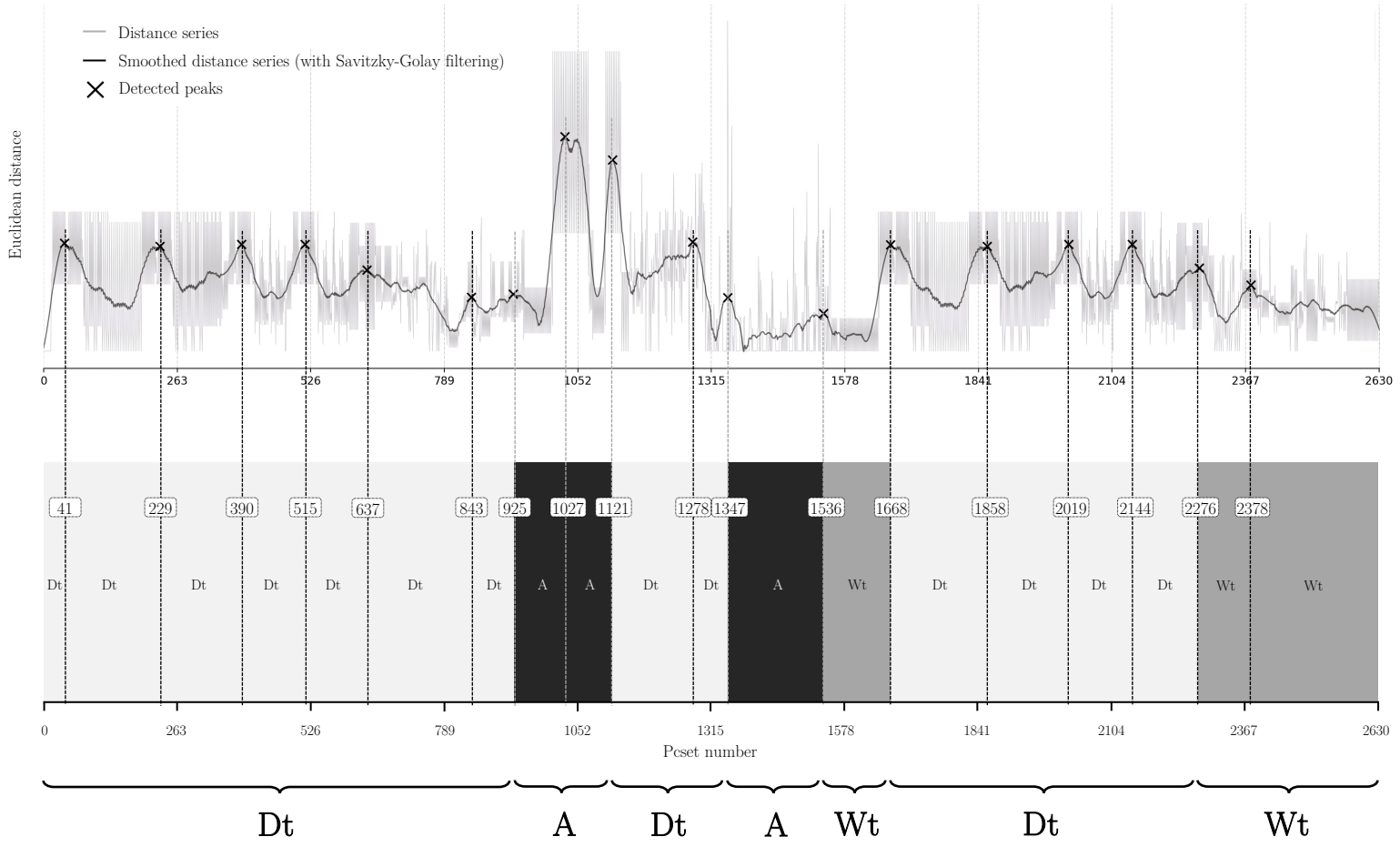
The distance between a point's neighbours is small inside a quality and large across a change of quality.

2.2. Adaptive segmentation: from peaks to a string.

1. Savitzky—Golay smoothing of the distance series, followed by prominence-based peak detection.

2. Recompute the DFT and the FQS region/quality for the respective segment.

Distance series, segmentation peaks, and qualia progression for Debussy's *Images*, L. 110: No. 3, *Mouvement*.

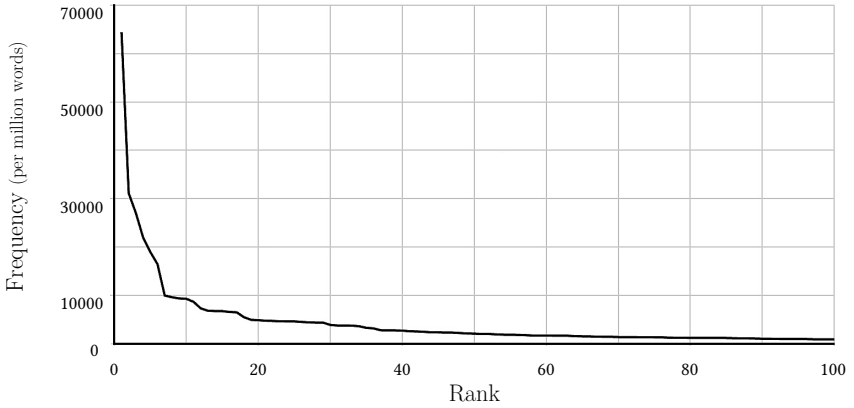


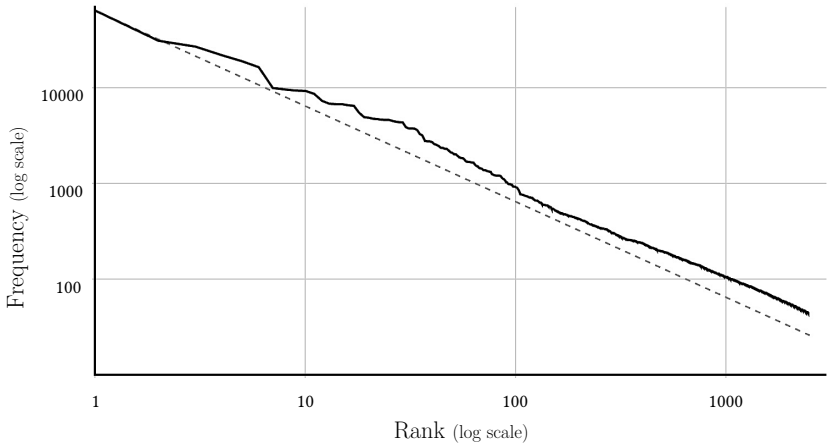
3.1. Zipf's law: the concept.

- $f(z) \propto 1/z^{\alpha}$ — the z -th most frequent token appears with a frequency proportional to $1/z^{\alpha}$.

- Natural language: $\alpha \approx -1$. The least-effort distribution.

Word	Rank	FrW
the	1	64420
of	2	31109
and	3	27002
a	4	21972
in	5	18978
to	6	16442
is	7	9961
to	8	9620
was	9	9368
it	10	9298
for	11	8664
that	12	7315
with	13	6821
he	14	6756
be	15	6742
on	16	6569
I	17	6494
by	18	5493
's	19	4945
at	20	4868
⋮	⋮	⋮



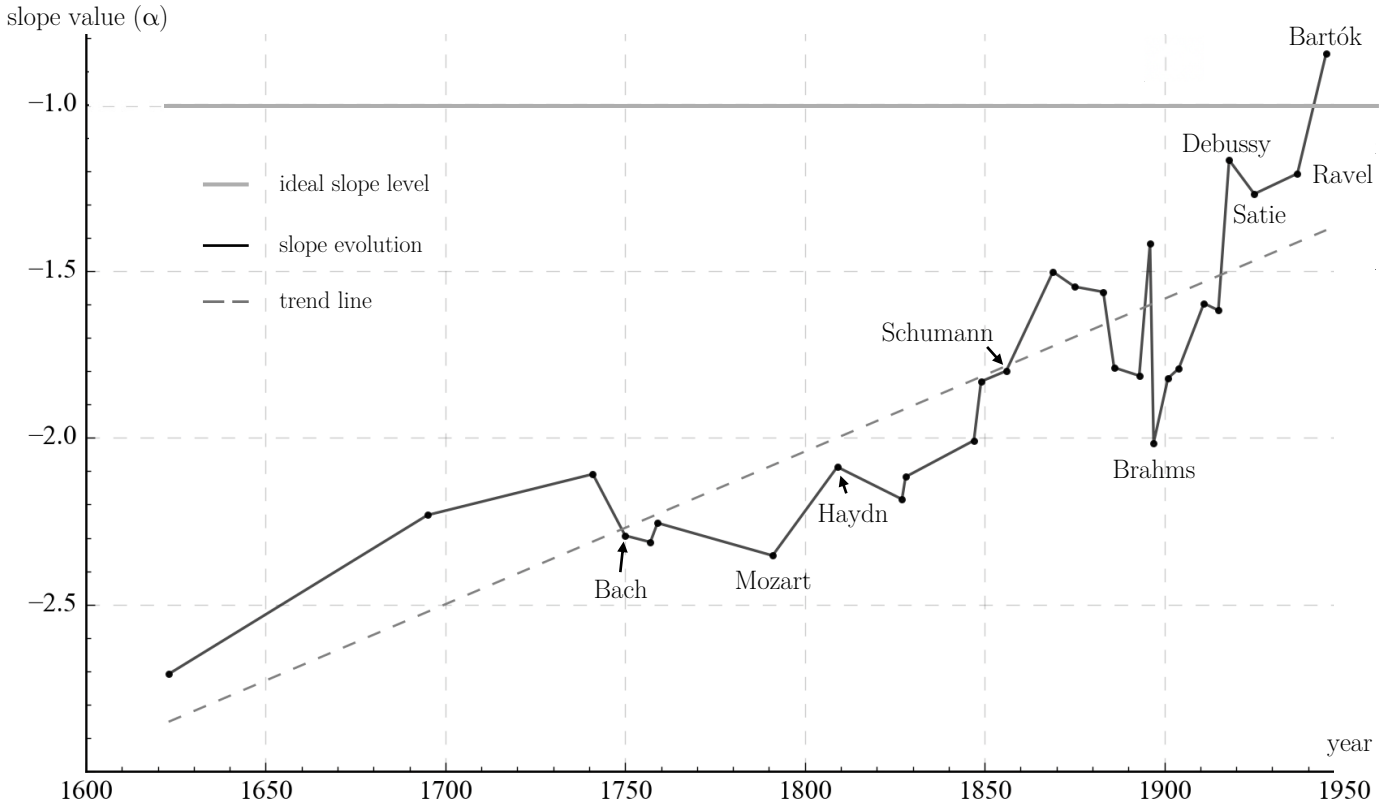


Word counts, and rank-frequency plots appearing on the written part of the *British National Corpus*.

3.2. Zipf's law: qualia bigrams become language-like.

Precedent: Rohrmeier & Cross (2008) that found Zipfian distributions governing chord succession in Bach's chorales.

- Harmonic qualia bigrams become progressively more language-like over four centuries.

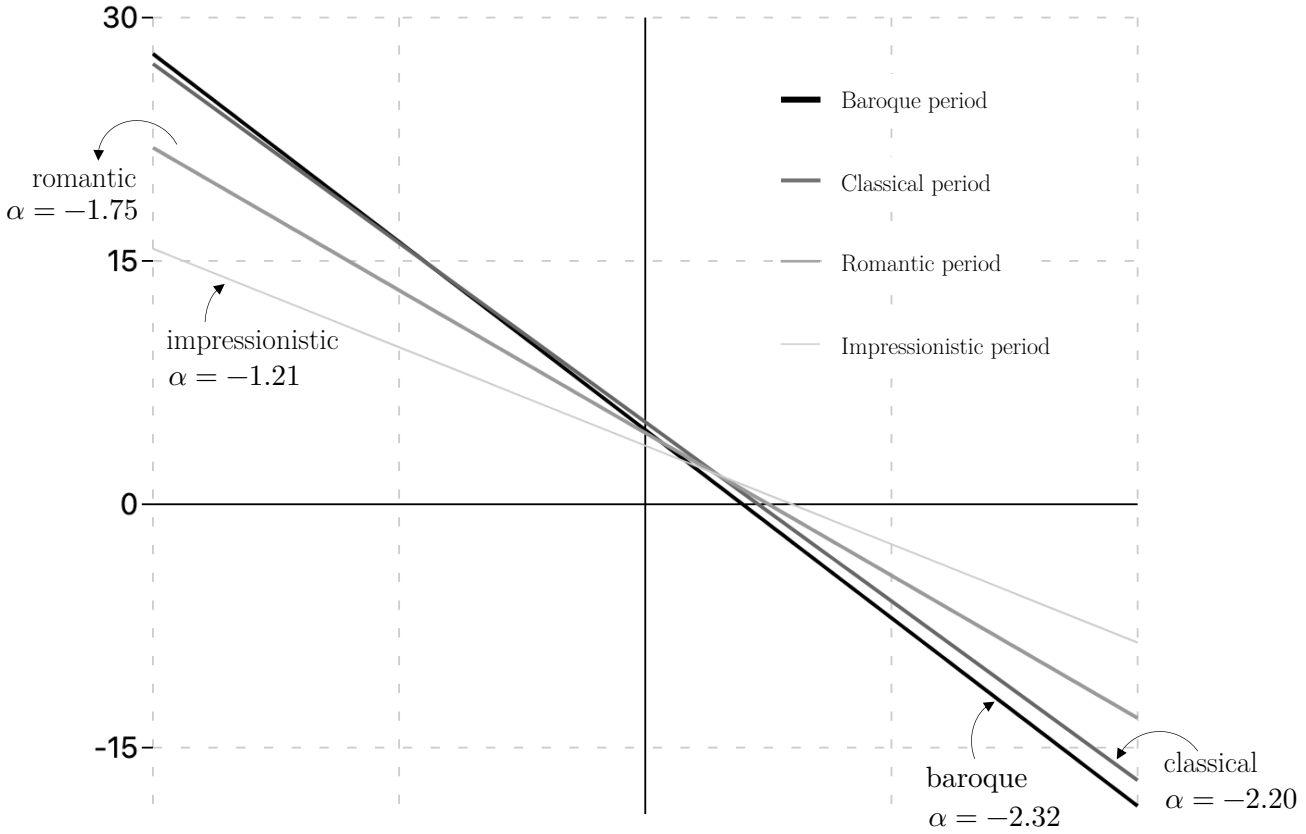


Per-composer log-rank/log-frequency slopes for qualia bigrams.

3.2. Zipf's mean slope by musical period.

- Four periods with one trajectory.
- The closer to $\alpha = -1$, the more language-like the qualia syntax.

	α	Composers
Baroque	-2.32	Byrd, Purcell, Vivaldi, Bach, Scarlatti, Handel
Classical	-2.21	Mozart, Haydn, Beethoven
Romantic	-1.76	Schubert through Mahler (14 composers)
Impressionist	-1.21	Debussy, Satie, Ravel



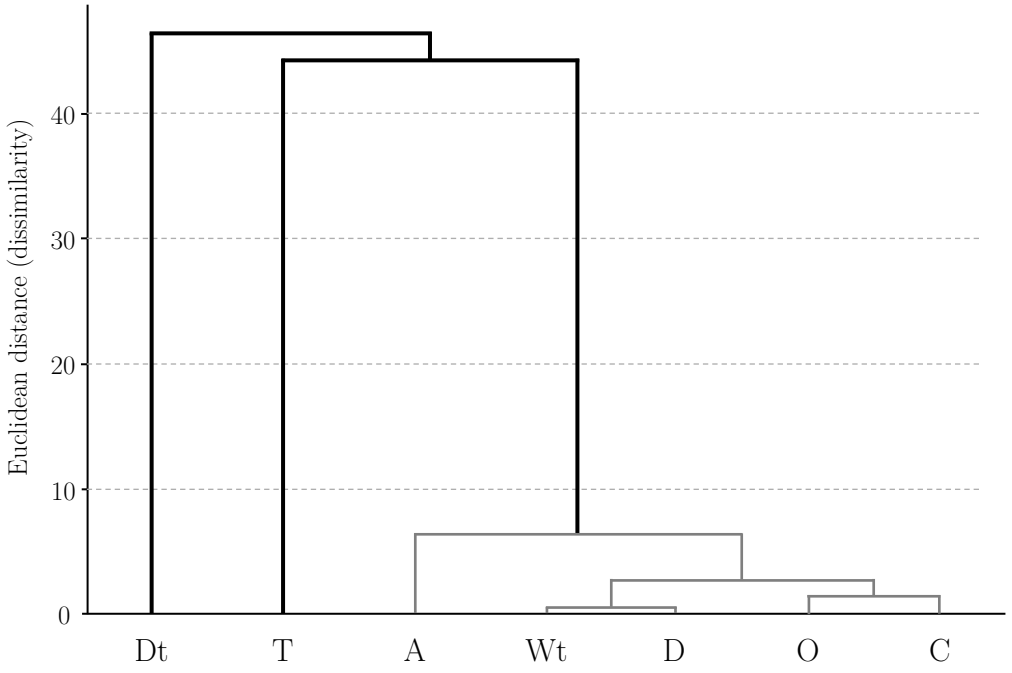
Period-aggregated regressions.

4. From bipartite to tripartite hierarchical clustering.

Hierarchical clustering on transition matrices (Ward's method).

Bach—bipartite

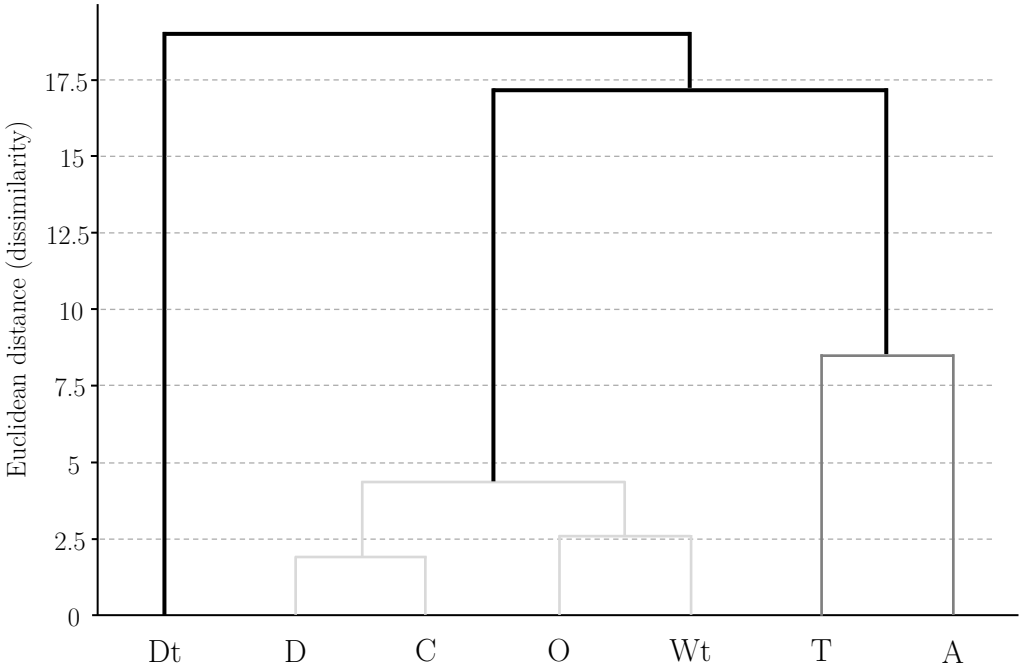
- Cluster 1: Dt + T (tonal core). Cluster 2: all others.



HCA dendrogram on Bach's transition matrices.

Debussy—tripartite

- Dt alone (structural isolation).
- T + A (mediating layer).
- O, Wt, C, D (colouristic periphery).



HCA dendrogram on Debussy's transition matrices.

5. Debussy's three-role model.

Anchor—Diatonicity

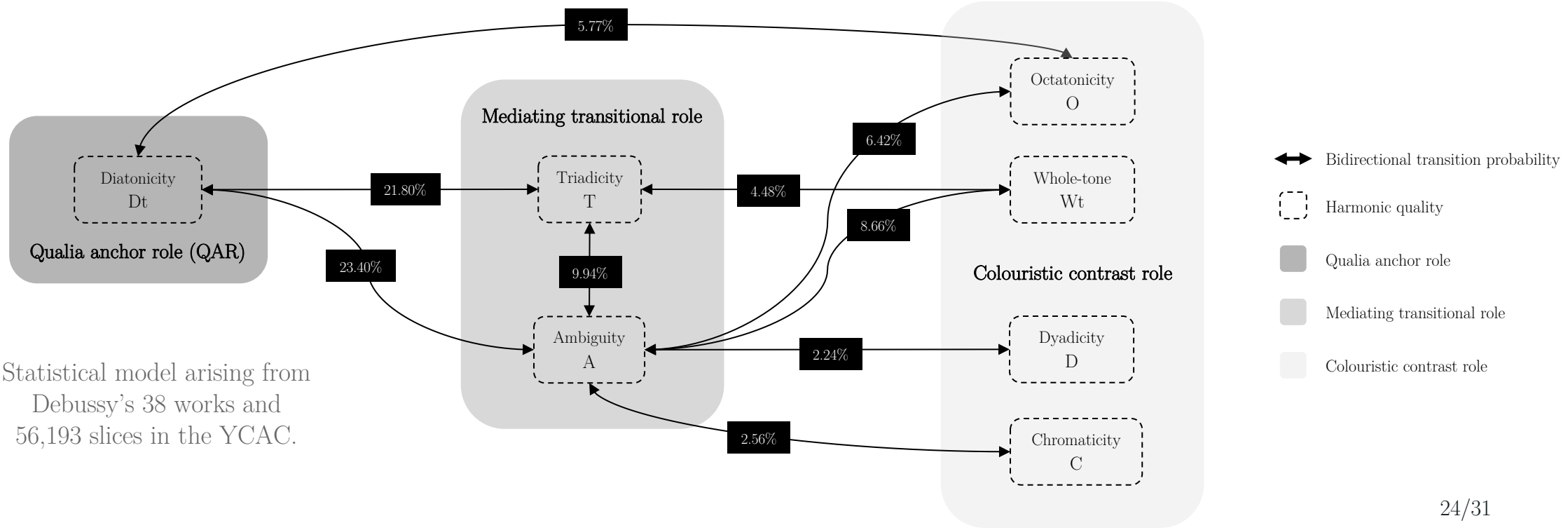
- 44.9% of transitions from Dt go to A; 61.3% from T return to Dt.

Mediating transitional—T, A

- Ambiguity as hub: 40.7% → Dt, 21% → T, 19.8% → Wt.

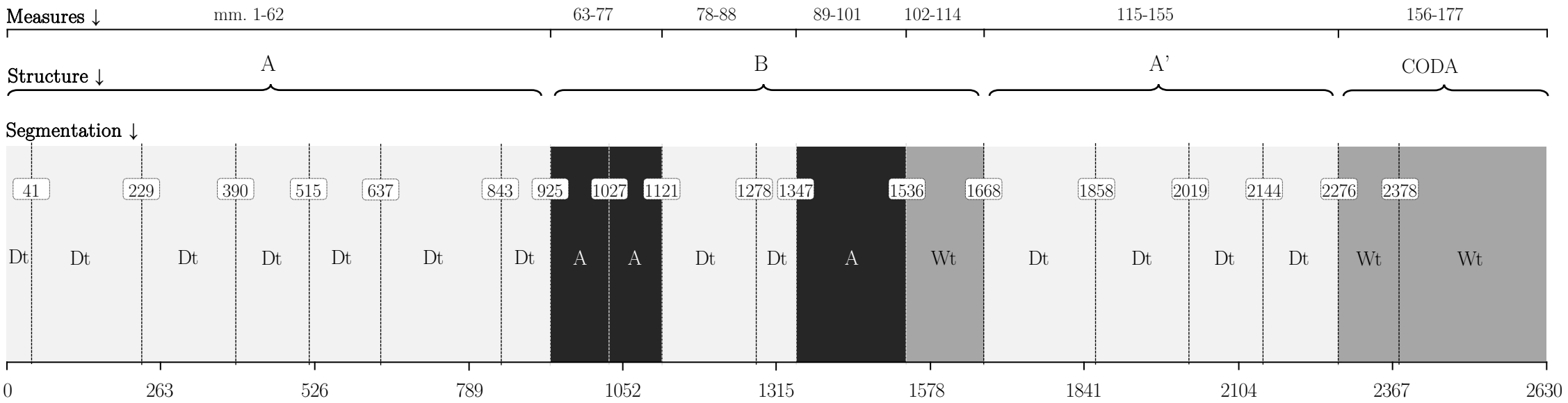
Colouristic contrast—O, Wt, C, D

- Rarely connect directly; 32.67% of trigrams are palindromic (Dt→A→Dt).



Statistical model arising from Debussy's 38 works and 56,193 slices in the YCAC.

6.1. *Mouvement*: a qualia progression that mirrors the form.



Roles ↓

QAR
Initial reference point;
Mainly major scales, pentatonic lines, and Lydian mode;
Short chromatic nuances.

MTR
Bridge into whole-tone quality;
Octatonic, chromatic, and whole-tone;
Brief interruption with diatonicity.

CCR
Contrast
Tension.

QAR
Initial reference point re-exposure;
Qualia tension resolution.

CCR
Final qualia contrast;
Unresolved tension.

Progression ↓
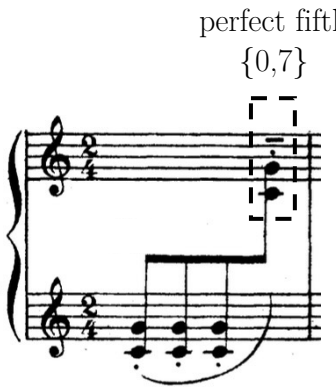
Dt → A → Dt → A → Wt → Dt → Wt →

Formal and qualia analyses across Debussy's *Images, Livre* 1: III. *Mouvement*.

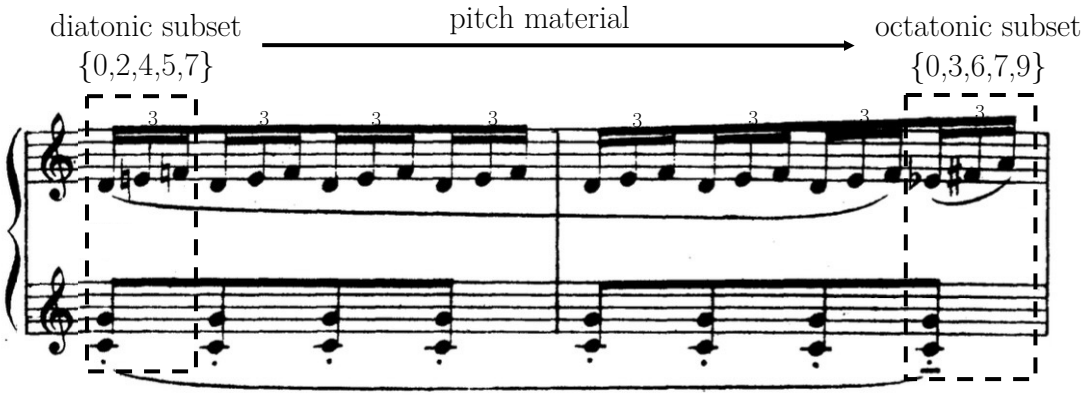
6.2. *Mouvement*: why the model captures it?

Diatonic {C,D,E,F,G}/octatonic subsets {C,E♭,F♯,G,A}. The interruption is brief but precisely placed under each tenuto.

m. 1 ↓



mm. 5-6 ↓



Articulation patterns: 1. 2. 3. 4__ 1. 2. 3. 4. 5. 6. 7. 8____

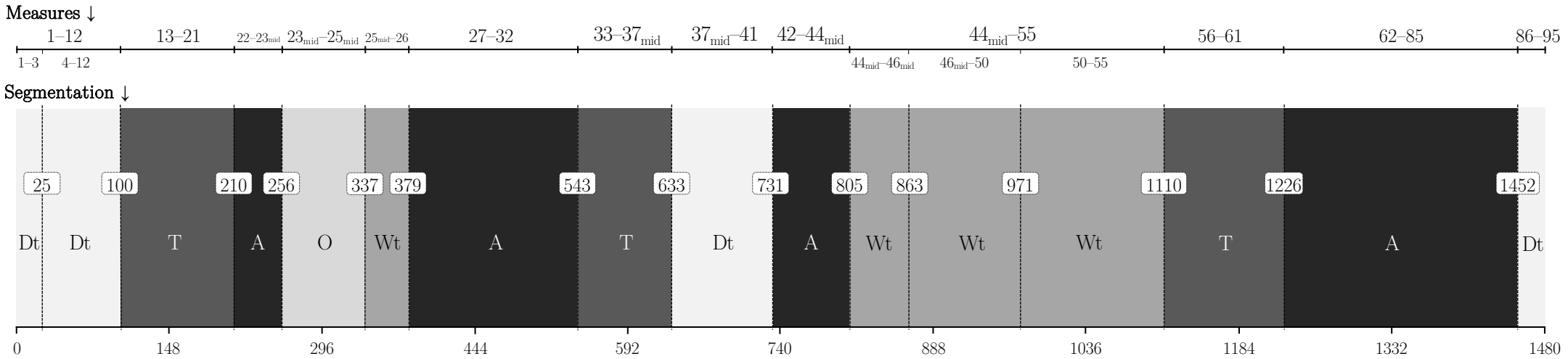
Articulation: 3+1, then 7+1. The pattern of an emphasised last element is mimicked by the qualia progression.

Listen.

Debussy, *Mouvement* mm. 1—41.

The rhythmic, articulative, and pitch organisation patterns creating the sense of "*movement*" in Debussy's Images, L. 110: No. 3.

7.1. *Reflets dans l'eau* through the model.



Progression ↓

Dt → T → A → O → Wt → A → T → Dt → A → Wt → T → A → Dt

Roles ↓

QAR
- First qualia anchor.
- Gravitational pull.

MTR
- Triadicity arising from joint octatonic, pentatonic, and triadic elements.
- Ambiguity defined by two chromatic melodic lines in contrary motion.

CCR
- Strongest departure from qualia stability.
- Introducing contrast and tension.

MTR
- Articulates diatonic, triadic, and chromatic elements.
- Facilitates the return to qualia anchor role.
- Transitional character emphasised by Tempo slowing down.

QAR
- Second qualia anchor.
- Resolution of qualia tension.
- Starts another cycle.

MTR
- New transition passage.
- Different context.
- Distinct role.

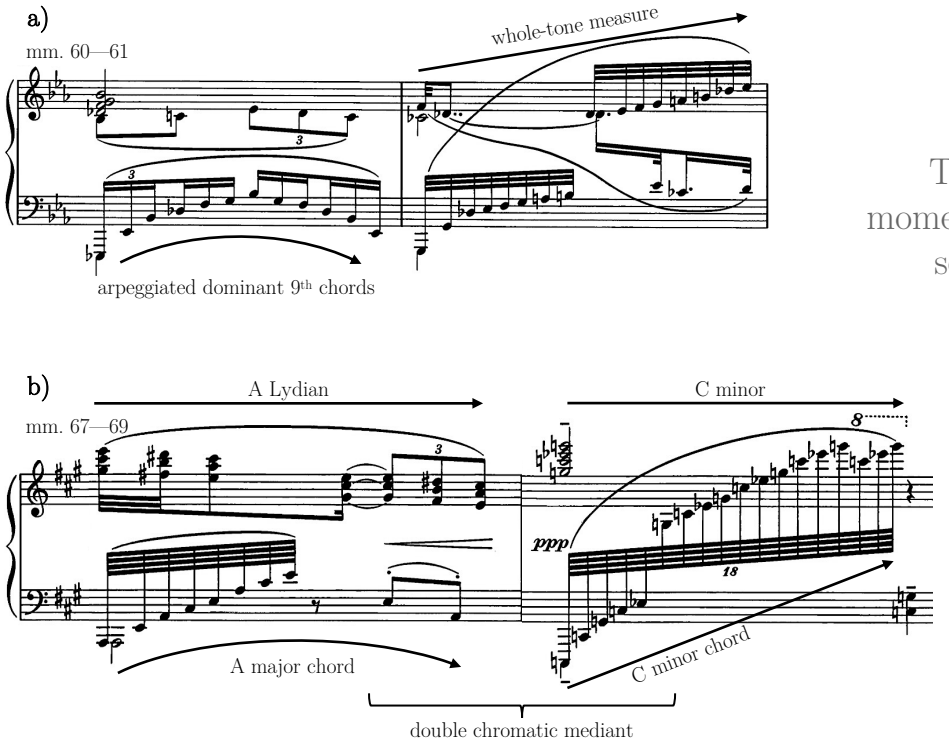
CCR
- Mainly whole-tone material;
- Second and final moment of colouristic contrast in the work.

MTR
- Work's climax (m. 57).
- Multiple harmonic shifts.
- Articulates diatonic, triadic, and chromatic elements.
- Facilitates the return to qualia anchor role.

QAR
- Final qualia rest.
- Slicing slow down.
- Final resolution of qualia tension.

7.2. Inside the mediating layer: *Reflets'* structured ambiguity.

- Inside the ambiguous segment, a concentration of qualia activity.



Two examples of qualia moments defining mm. 62—85's segment as ambiguous.

Listen.

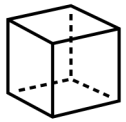
Debussy, *Reflets dans l'eau* mm. 60—70.
Listen 2/2.

V. Postlude.

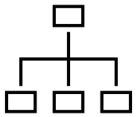
Conclusion.

1. What the framework adds, and where it falters.

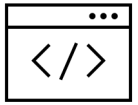
Contributions:



Methodological: a six-dimensional DFT projection rendered as a single hexagonal layout.



Theoretical: anchor, mediating, and colouristic roles recovered statistically.



Technical: adaptive distance-sensitive segmentation.

Limitations:

- Purely harmonic: rhythm, timbre and form lie outside the model.
- Emergent regularities, not documented compositional principles. No claim about intention.

What comes next:

- Bartók, Ravel, Stravinsky, Messiaen...
- Grammar induction over qualia strings; perceptual testing.
- Tonfeld theory.

Thank you! :)

Selected references:

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Mailman, Joshua. 2015. "Schoenberg's Chordal Experimentalism Revealed through Representational Hierarchy Association (RHA), Contour Motives, and Binary State Switching." Music Theory Spectrum 37 (2): 224–52.

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Rohrmeier, Martin and Ian Cross. 2008. "Statistical Properties of Tonal Harmony in Bach's Chorales". In Proceedings of the 10th International Conference on Music Perception and Cognition (ICMPC).

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